

TRANSFORMERS

Complete Toy Collection Tracker

Technical Documentation

Version	1.0 — May 2026
File	index.html (single-file application)
Hosted at	bogus13pt.github.io/transformers/
Entries	924 toy figures
Toy Lines / Eras	58 eras
Images	924 JPG files in imgs/
Limited Editions	37 figures with production data
File Size	~170KB HTML + ~250MB images
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This document is a complete technical reference. It covers architecture, data model, all JavaScript functions, CSS design system, deployment, and maintenance procedures. Intended to allow full recovery and continued development without access to the original development session.

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1. Project Overview

The Transformers Collection Tracker is a **self-contained single-file HTML application** that tracks ownership of Transformers toy figures across all major toy lines from 1980 to 2026. It requires no backend, no database, and no build process.

Key Design Decisions

- **Single-file architecture:** HTML, CSS, and JavaScript in one file. Images in imgs/. Works offline.
- **No framework:** Pure vanilla JavaScript. No React, Vue, or jQuery. No dependency rot.
- **localStorage persistence:** Owned figures stored in browser localStorage under key tf_owned.
- **GitHub Pages hosting:** Repository bogus13pt/transformers at bogus13pt.github.io/transformers/
- **Static images:** 924 JPG images (~250MB) in imgs/ alongside index.html.

Technology Stack

Layer	Technology	Notes
Markup	HTML5	Single index.html
Styling	CSS3 custom	Variables, animations, responsive grid
Logic	Vanilla JavaScript ES6	No frameworks or dependencies
Fonts	Google Fonts	Orbitron, Share Tech Mono, Rajdhani
Storage	localStorage	Owned state persisted in browser
Hosting	GitHub Pages	bogus13pt/transformers repository
Images	JPEG 200-400KB each	924 files in imgs/ folder

2. File Structure

The repository has a flat structure with the HTML file at the root and all images in imgs/:

```
bogus13pt/transformers (GitHub repository) |-- index.html <- Main application file |--  
.nojekyll <- Disables Jekyll on GitHub Pages `-- imgs/ <- 924 JPG image files |--  
g1-optimus-prime.jpg |-- wst-optimus.jpg |-- ld-gold-convoy-g1.jpg `-- ... (924 total)
```

Internal HTML Structure

```
index.html |-- <head> | |-- Meta tags (charset, viewport) | |-- Google Fonts links | `--  
<style> all CSS (~400 lines) `-- <body> |-- LIGHTBOX HTML (must be FIRST, before script) | |--  
<div id="lb-overlay"> | |-- <button id="lb-close-btn"> | |-- <button id="lb-prev-btn"> | `--  
<button id="lb-next-btn"> |-- <header> title + stats bar |-- <div id="toolbar"> search +  
filters + export/import |-- <div id="output"> card grid (built by JS) |-- <footer> legal  
disclaimer `-- <script> all JavaScript (~1300 lines) |-- const toys = [...] 924 toy entries  
|-- const eraOrder, eraColors, eraYears |-- const factionColors, allClasses, classColors |--  
lbOpen(), lbClose(), lbShow() |-- makeCard() |-- buildFilterBar() |-- render() |--  
updateStats() `-- Event listeners
```

CRITICAL: The lightbox HTML elements MUST appear in the DOM before the script tag. If placed after the script, `getElementById()` returns null and throws `TypeError` on page load.

3. Data Model -- Toy Entry Format

Each figure is a JavaScript object in the `const toys = [...]` array:

```
{ era: "G1", // required faction: "Autobot", // required cls: "Car", // required name:
  "Optimus Prime", // required note: "Leader/Commander", // required col: "#e63946", // required
  -- hex accent colour img: "imgs/g1-optimus-prime.jpg", // required -- relative path //
  Optional: units: "~17,000 units", // limited edition badge year: "2021", // explicit year
  override }
```

Field Reference

Field	Req	Description
era	Yes	Must match key in eraOrder. Groups cards by toy line.
faction	Yes	Autobot / Decepticon / Maximal / Predacon / Vehicon / Rebel / Empire / Republic / Bounty
cls	Yes	Car / Jet / Air / Beast / Combiner / Headmaster / Pretender / Cassette / Exclusive / MP / AM / Mini / Vintage
name	Yes	Display name. Shown in Orbitron font on card.
note	Yes	Short description. Year stripped from display if shown in year badge.
col	Yes	CSS hex colour for card accent, class tag tint, placeholder text.
img	Yes	Path relative to index.html. Always imgs/filename.jpg.
units	No	Production run. Triggers gold border + crown + LIMITED badge.
year	No	Explicit year. Overrides auto-extraction from note and eraYears.

Adding a New Entry

```
{era:"Legacy United", faction:"Autobot", cls:"Car", name:"New Figure", note:"Deluxe Class",
col:"#e63946", img:"imgs/lu-new-figure.jpg"},
```

4. Complete Era List (58 eras)

Eras are sorted chronologically. The eraOrder array defines display order.

#	Era / Toy Line	Year	Count	Colour
1	Pre-TF Diaclone	1980	10	#f39c12
2	Pre-TF Micro Change	1983	6	#7f8c8d
3	G1	1984	112	#e63946
4	Lucky Draw	1984	25	#c0a000
5	Action Masters	1990	13	#e74c3c
6	G2	1992	14	#f97316
7	Beast Wars	1996	21	#22c55e
8	Machine Wars	1997	8	#7f8c8d
9	Beast Machines	1999	9	#16a34a
10	RID 2001	2001	9	#3b82f6
11	Armada	2002	12	#8b5cf6
12	World's Smallest	2002	27	#e74c3c
13	Alternators	2003	15	#3498db
14	Binaltech	2003	5	#c0392b
15	Masterpiece	2003	40	#c0a000
16	Takara STF	2003	12	#8e44ad
17	Energon	2004	13	#06b6d4
18	Cybertron	2005	11	#0891b2
19	SW Transformers	2005	24	#c8a45a
20	Classics	2006	18	#f59e0b
21	Movie 2007	2007	15	#f97316
22	Animated	2008	20	#e63946
23	Henkei	2008	11	#3498db
24	Marvel Crossovers	2008	7	#e74c3c
25	Shattered Glass	2008	9	#8e44ad
26	ROTF	2009	13	#6366f1
27	Generations	2010	14	#22c55e
28	DOTM	2011	15	#64748b
29	Rescue Bots	2012	8	#f97316
30	TF Prime	2012	14	#06b6d4
31	AOE	2014	13	#f59e0b
32	Angry Birds TF	2014	6	#e74c3c
33	Combiner Wars	2014	13	#e74c3c
34	Micro Figure & Comic	2014	5	#c0392b
35	Minimates	2014	5	#f39c12

36	Keshi Surprise	2015	6	#27ae60
37	RiD 2015	2015	8	#22c55e
38	Titans Return	2016	15	#0891b2
39	Micro Toy Box	2017	6	#2980b9
40	Studio Series	2017	41	#ffd60a
41	TLK	2017	10	#a855f7
42	Cyberverse	2018	8	#f97316
43	POTP	2018	31	#ec4899
44	Vintage G1 Reissue	2018	9	#f59e0b
45	WFC: Siege	2018	21	#ef4444
46	Collaborative	2019	41	#ff6b9d
47	Gen Selects	2019	8	#22c55e
48	HasLab	2020	4	#ffd60a
49	R.E.D.	2020	7	#ec4899
50	WFC: Earthrise	2020	17	#f59e0b
51	WFC: Kingdom	2021	19	#22c55e
52	Blokees	2022	7	#0891b2
53	EarthSpark	2022	10	#06b6d4
54	Legacy	2022	21	#3b82f6
55	Legacy Evolution	2023	26	#06b6d4
56	ROTB	2023	11	#e63946
57	Age of Primes	2024	10	#c0a000
58	Legacy United	2024	16	#3b82f6

5. JavaScript Architecture

Single inline script block. No modules. Synchronous and event-driven.

```
Script execution order: 1. const toys = [...] Data array (924 entries) 2. Lookup tables
eraOrder, eraColors, eraYears, factionColors, allClasses 3. State variables owned (Set),
currentEra, currentFaction, viewMode 4. lbOpen/lbClose/lbShow Lightbox (defined BEFORE
makeCard) 5. makeCard(data, delay) DOM factory for card elements 6. buildFilterBar(...) Filter
pill buttons 7. render() Main render loop 8. updateStats() Header counters 9. setView(mode)
Grid vs list toggle 10. Init buildFilterBar x3, updateStats(), render() 11. Event listeners
search, export, import, keyboard
```

State Variables

```
const owned = new Set(JSON.parse(localStorage.getItem("tf_owned") || "[]")); let currentEra =
"All"; let currentFaction = "All"; let currentClass = "All"; let viewMode = "grid"; var
_lbList = []; // lightbox navigation list var _lbIdx = 0; // current lightbox index
```

Ownership Key

```
function itemKey(d) { return d.era + "|" + d.name + "|" + d.note; } // Example: "G1|Optimus
Prime|Leader/Commander" // Stored as JSON array in localStorage["tf_owned"]
```


6. Key Functions Reference

Card DOM Factory

`makeCard(data, delay)`

Creates a complete card DOM element for one toy figure. Called by `render()` per toy.

```
1. Create .card div with era colour border 2. Faction badge (AUTOBOT / DECEPTICON / etc.) 3.
.card-img area with placeholder triangle 4. img element: cursor=zoom-in, click -> lbOpen() 5.
Owned button: toggles owned Set + localStorage 6. yearVal = data.year || noteYear(regex) ||
eraYears[era] 7. cleanNote: strip trailing year from note text 8. body.innerHTML: era / name /
tags / note 9. If data.units: gold border + crown + LIMITED badge 10. Return card element
```

Main Render Loop

`render()`

Filters toys[], groups by era, builds DOM. Called on load, filter change, search.

```
1. Read search query from #search 2. Filter: currentEra AND currentFaction AND currentClass
AND search text (search checks name + era + faction + cls + note, case-insensitive) 3. Group
results by era (eraOrder sequence) 4. Per era: section header + grid div + makeCard() per toy
5. Update #countPill ("Showing N items") 6. window._lbCurrentList = filtered 7. updateStats()
```

Filter Button Builder

`buildFilterBar(id, items, colorFn, labelFn, setter)`

Generic builder for ERA, FACTION, CLASS filter pill rows.

```
id -- DOM container id items -- array of filter values colorFn -- function(item) -> hex colour
labelFn -- function(item) -> display text setter -- function(val) called on click, then
render() Always prepends ALL button. Highlights active selection.
```

Lightbox Open

`lbOpen(data)`

Opens image zoom overlay. Sets navigation list from current filtered toys.

```
1. _lbList = window._lbCurrentList 2. Find index of clicked toy in _lbList 3. lbShow() to
populate image + caption 4. Add .open class to #lb-overlay 5. Set display:flex on
close/prev/next buttons
```

Lightbox Close

`lbClose()`

Closes overlay and hides buttons. Triggered by X click, outside click, Escape key.

```
1. Remove .open from #lb-overlay 2. Set display:none on close/prev/next buttons
```

Lightbox Show

`lbShow()`

Updates image and caption to `_lbList[_lbIdx]`. Called by `lbOpen` and nav buttons.

```
1. d = _lbList[_lbIdx] 2. #lb-img.src = d.img 3. #lb-title = d.name 4. #lb-sub = era + faction
+ note Arrow keys: increment/decrement _lbIdx modulo length
```

Header Counters

updateStats()

Updates owned/total/percent/years counters in page header.

```
#ownedCount = owned.size #totalCount = toys.length #pctCount = Math.round(owned.size /  
toys.length * 100) #yearsCount = "40+" (static)
```

7. CSS Architecture

All CSS in a single style block in head. Dark sci-fi design language.

Colour Variables

Variable	Value	Usage
--bg	#05080c	Page background
--surface	#080c12	Card background
--red	#e63946	Primary accent (Autobot)
--gold	#ffd60a	Limited edition / year badge
--green	#2ec4b6	Owned state
--white	#e8edf2	Primary text
--muted	rgba(200,214,229,.45)	Secondary text

Key CSS Sections

- **Card Grid:** CSS Grid, auto-fill, minmax(180px, 1fr), gap 16px
- **Card:** dark surface, era-colour border, hover lift animation, rise keyframe
- **Card Image Area (.card-img):** fixed height 210px, overflow:hidden, placeholder triangle
- **Card Body:** .card-era (mono font), .card-name (Orbitron), .class-tag, .year-tag (gold), .card-note
- **Limited Edition:** gold shimmer border animation, goldShimmer keyframe, .limited-badge inline
- **Lightbox:** position:fixed, display:none / display:flex (.open), dark overlay, blur
- **Lightbox Buttons:** position:fixed, circular, display:none by default -- shown via JS only

8. Lightbox / Zoom System

Full-size image zoom overlay with keyboard navigation.

HTML Structure

```
<!-- Right after <body>, BEFORE <header> and <script> --> <div id="lb-overlay"> overlay  
(display:none default) <div id="lb-box"> centred content <img id="lb-img"> zoomed image <div  
id="lb-caption"> name + details <strong id="lb-title"> <span id="lb-sub"> <button  
id="lb-close-btn">X top-right, position:fixed, display:none <button id="lb-prev-btn"> left  
edge, position:fixed, display:none <button id="lb-next-btn"> right edge, position:fixed,  
display:none
```

Critical Implementation Notes

- **DOM order:** Lightbox HTML MUST be before the script tag. Otherwise `getElementById()` returns null.
- **Button visibility:** Buttons use `inline style="display:none"`. Controlled via JS only -- never via CSS classes. Prevents `position:fixed` elements from escaping parent visibility.
- **Overlay toggle:** CSS class `.open` switches `display:none` to `display:flex` on `#lb-overlay`.
- **Navigation list:** `lbOpen()` reads `window._lbCurrentList` set by `render()`. Nav only cycles visible filtered figures.

9. Export / Import System

Backup and restore collection data across browsers or devices.

Export JSON Format

```
{ "version": 1, "app": "Transformers Collection Tracker", "exported":  
  "2026-05-12T09:50:38.000Z", "owned": [ "G1|Optimus Prime|Leader/Commander",  
  "G1|Bumblebee|Mini-vehicle", ... ] } // File name: transformers-collection-YYYY-MM-DD.json
```

Import Logic

```
// Accepts both new format (with wrapper) and legacy (plain array) const keys =  
Array.isArray(data.owned) ? data.owned : Array.isArray(data) ? data : null; owned.clear();  
keys.forEach(k => owned.add(k)); localStorage.setItem("tf_owned",  
JSON.stringify([...owned])); render();
```

10. Limited Edition System

There are currently **37 limited edition figures**. Triggered by the units field.

Visual Treatment

- **Gold shimmer border:** .limited class -- goldShimmer CSS animation pulses box-shadow
- **Crown icon:** appended to .card-img at top-centre
- **LIMITED badge:** inline in .card-body below note text (not position:absolute)

Example Entries

```
{era:"HasLab", name:"Unicron", units:"~17,000 units"}, {era:"Lucky Draw", name:"Mayuko Iwasa Meister", year:"2006", units:"1 unit"}, {era:"Shattered Glass", name:"Optimus Prime (SG)", year:"2008", units:"~1,500 units"}
```

11. Year Tag System

Three-level priority system for year display:

```
Priority 1: data.year field (explicit override) Priority 2: year extracted from note via regex
/\b(19[89]\d|20[012]\d)\b/ Priority 3: eraYears[era] default lookup Note text cleaning
(prevent year appearing twice): .replace(/\s*(YYYY\)/g, "") removes (1982) .replace(/\s*--
YYYY$/g, "") removes -- 2003 at end .replace(/\s+YYYY$/g, "") removes trailing year Examples:
"Diaclone -> Wheeljack (1982)" -> "Diaclone -> Wheeljack" (year to badge) "BotCon 2008 -- evil
Autobot" -> unchanged (year mid-string)
```

12. Filter and Search System

Filter Logic

```
const filtered = toys.filter(d => { if (currentEra !== "All" && d.era !== currentEra) return false; if (currentFaction !== "All" && d.faction !== currentFaction) return false; if (currentClass !== "All" && d.cls !== currentClass) return false; if (q) { const hay = (d.name+d.era+d.faction+d.cls+d.note).toLowerCase(); if (!hay.includes(q)) return false; } return true; }); // Filters are AND-combined. All conditions must pass.
```

Filter Bars

```
buildFilterBar("eraBar", eraOrder, era => eraColors[era], era => era, ...);
buildFilterBar("factionBar", allFactions, f => factionColors[f], f => f, ...);
buildFilterBar("classBar", allClasses, cls => classColors[cls], cls => cls, ...);
```


13. GitHub Pages Deployment

Repository Settings

```
Repository : bogus13pt/transformers Branch : main Folder : / (root) Live URL :  
https://bogus13pt.github.io/transformers/ Required files: index.html <- application .nojekyll  
<- empty file, disables Jekyll imgs/ <- all 924 JPG files
```

Critical Requirements

- **.nojekyll**: Must exist at root. Without it, Jekyll processes the HTML and 0 toys may show.
- **index.html**: Must be named index.html (not transformers.html).
- **Image paths**: All img fields use relative paths (imgs/file.jpg). index.html and imgs/ must be in same folder.
- **Cache**: After upload, force refresh with Ctrl+Shift+R. Image CDN cache clears in 10-60 minutes.
- **Upload**: Use "Add file -> Upload files" in GitHub web UI. Avoid web editor for large files.

Deployment Checklist

```
[ ] index.html uploaded to repository root [ ] .nojekyll file present at root [ ] imgs/ folder  
with all JPG files [ ] GitHub Pages: Settings -> Pages -> Branch: main, Folder: / [ ] Wait ~2  
minutes for GitHub Actions deployment [ ] Hard refresh: Ctrl+Shift+R
```

14. Maintenance and Adding New Toys

Adding a New Era

```
// Step 1: Add entries to toys array {era:"New Era", faction:"Autobot", cls:"Car",  
name:"Figure", note:"Desc", col:"#hex", img:"imgs/ne-figure.jpg"}, // Step 2: Add to eraOrder  
(chronological position) const eraOrder = [..., "New Era", ...]; // Step 3: Add to eraColors  
const eraColors = {..., "New Era": "#hexcolour"}; // Step 4: Add to eraYears const eraYears =  
{..., "New Era": "2025"}; // Step 5: Upload image to imgs/ on GitHub
```

Common Issues and Solutions

Page loads but shows 0 toys

JavaScript syntax error in toys array. Open DevTools (F12) -> Console. Look for SyntaxError. Common cause: unescaped double-quote inside note or name. Fix: replace with escaped version.

Image shows placeholder triangle

Image file missing from imgs/ on GitHub, or filename mismatch. Filenames are case-sensitive. Check data img field vs actual filename.

Year badge not showing

eraYears has no entry for that era, data.year is missing, and note has no 4-digit year. Fix: add year:"YYYY" to the toy entry.

Lightbox buttons always visible on page

The lb-close-btn, lb-prev-btn, lb-next-btn elements are after the script tag. Move lightbox HTML to immediately after body.

Multiple images look identical

Fallback mechanism copied one file to all slots during failed downloads. Re-download: check file hashes, re-run with more specific queries.

Deployed page not updating after upload

GitHub Pages CDN cache. Force browser refresh with Ctrl+Shift+R. Wait 10-60 minutes for full CDN propagation.